

SERENA R. SCHOLZ

Department of Earth and Planetary Sciences, Yale University
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EDUCATION

Ph.D., Earth and Planetary Sciences, Yale University 2022–present

- Atmosphere, Ocean and Climate Dynamics Division; expected defense May 2027.
- Committee: Juan Lora (advisor), Alexey Fedorov, Mary-Louise Timmermans, John Wettlaufer.

M.S., Earth and Environmental Sciences, University of Michigan 2020

- Advisor: Sierra Petersen.

B.S., Earth and Environmental Sciences, University of Michigan 2019

EXPERIENCE

Associate Research Analyst, CNA Institute for Public Research 2020–2022

- Research and analysis for federal and state government clients on natural disasters and emergency response, infrastructure resilience, cybersecurity, and supply chain disruptions.

Research Assistant (part-time), Lamont-Doherty Earth Observatory 2019–2021

- Studied centennial-scale Euro-Mediterranean hydroclimate variability in the Ocean & Climate Physics group. Advisor: Richard Seager.

Research Assistant (part-time), University of Michigan 2016–2020

- High-resolution oxygen and clumped isotope sclerochronology for paleoclimate reconstructions. Advisor: Sierra Petersen

PUBLICATIONS

1. **Scholz, S.R.** & J.M. Lora (2025). Widespread increase in atmospheric river frequency and impacts over the 20th century. *AGU Advances* 6, e2025AV001888. doi:10.1029/2025AV001888
Eos Editor's Highlight.
2. **Scholz, S.R.** & J.M. Lora (2024). Atmospheric rivers cause warm winters and extreme heat events. *Nature* 636, 640–646. doi:10.1038/s41586-024-08238-7
Covered by Newsweek, Physics Today, Yale News, Yale Scientific, & others.
3. **Scholz, S.R.**, S.V. Petersen & B.M. Anderson (2024). Modern reconstructions of mean and seasonal-scale climate from coastal marine gastropods (Turritellidae). *Palaeogeography, Palaeoclimatology, Palaeoecology* 655, 112553. doi:10.1016/j.palaeo.2024.112553
4. **Scholz, S.R.**, R. Seager, M. Ting, Y. Kushnir, J.E. Smerdon, B.I. Cook, E.R. Cook & S.H. Baek (2022). Changing hydroclimate dynamics and the 19th to 20th century wetting trend in the English Channel region of northwest Europe. *Climate Dynamics* 58, 1539–1553. doi:10.1007/s00382-021-05977-5
5. O'Hara, H.E., S.V. Petersen, J. Vellekoop, M.M. Jones & **S.R. Scholz** (2022). Clumped-isotope-derived climate trends leading up to the end-Cretaceous mass extinction in northwestern Europe. *Climate of the Past* 18, 1963–1982. doi:10.5194/cp-18-1963-2022
6. Baek, S.H., J.E. Smerdon, G.-C. Dobrin, J.G. Naimark, E.R. Cook, B.I. Cook, R. Seager, M.A. Cane & **S.R. Scholz** (2020). A quantitative hydroclimatic context for the European Great Famine of 1315–1317. *Communications Earth & Environment* 1, 19. doi:10.1038/s43247-020-00016-3
Covered by State of the Planet, Columbia Climate School.
7. **Scholz, S.R.**, S.V. Petersen, J.H. Escobar, C. Jaramillo, A. Hendy, W.D. Allmon, J.H. Curtis, B.M. Anderson, N. Hoyos, J.C. Restrepo & N. Perez (2020). Isotope sclerochronology indicates enhanced seasonal precipitation in northern South America (Colombia) during the Mid-Miocene Climatic Optimum. *Geology* 48, 668–672. doi:10.1130/G47235.1

IN REVIEW	Scholz, S.R. & J.M. Lora. The dynamic lifecycle of atmospheric rivers: insights from idealized models. In review, <i>Science Advances</i> .
	O'Brien, T.A., K.T. Quagrain, C. Shields, S.R. Scholz & 35 others. Lessons from ten years of the Atmospheric River Tracking Method Intercomparison Project. In review, <i>Bulletin of the American Meteorological Society</i> .
	Mattingly, K.S., M. MacLennan, J. Schnaubelt, C. Shields, R. Baiman, A. Massa, K. Nardi, S.R. Scholz & 26 others. The ARTMIP Tier 2 polar synthesis: a comparative analysis of polar atmospheric river tracking methods in historical and future climate states. Submitted, <i>The Cryosphere</i> .
IN PREP.	Scholz, S.R. & J.M. Lora. Atmospheric river circulation and precipitation across climate states. In preparation, <i>Geophysical Research Letters</i> .
	Scholz, S.R. & A.V. Fedorov. Contrasting responses of the tropical Indian and Pacific Oceans to an anomalous surface heat flux. In preparation, <i>Geophysical Research Letters</i> .
OTHER WRITING	Scholz, S.R. & J.M. Lora (2026). Quick Study: Rivers in the Sky. <i>Physics Today</i> 79 (9), 44-46. doi:10.1063/pt.171e3ee933
	Scholz, S.R. , R. Composto, K. Inks, Z. Dutton, & L. Abraham (2021). Impacts of Climate Change on Supply Chains. White paper, CNA.
INVITED TALKS	Climate Dynamics and Impacts Seminar, Princeton University. "Atmospheric rivers cause warm winters and extreme heat events." March 2025.
PRESENTATIONS	Scholz, S.R. & J.M. Lora. Atmospheric rivers and energy transport in a hierarchy of idealized models. <i>EGU General Assembly</i> , Vienna, Austria, May 2026. (talk)
	Scholz, S.R. & J.M. Lora. Atmospheric rivers in a hierarchy of idealized models. <i>AMS Annual Meeting</i> , Conference on Climate Variability and Change, Houston, TX, January 2026. (talk)
	Scholz, S.R. & J.M. Lora. Atmospheric river structure and energy transport in a hierarchy of idealized models. <i>AGU Fall Meeting</i> , New Orleans, LA, December 2025. (talk)
	Scholz, S.R. & J.M. Lora. Global trends in atmospheric river temperatures, snowfall, and precipitation. <i>AGU Fall Meeting</i> , Washington, DC, December 2024. (talk)
	Scholz, S.R. & J.M. Lora. Global impacts of atmospheric rivers on surface temperatures and heat fluxes. <i>International Atmospheric Rivers Conference</i> , La Jolla, CA, June 2024. (talk)
	Scholz, S.R. & J.M. Lora. Atmospheric rivers cause extreme heat excursions and anomalously warm temperatures. <i>AGU Fall Meeting</i> , San Francisco, CA, December 2023. (talk)
	Scholz, S.R. & J.M. Lora. How hot are atmospheric rivers? <i>Yale Climate Day</i> , New Haven, CT, May 2023. (poster)
	Scholz, S.R. & S.V. Petersen. Challenges and advances in reconstructing mean and seasonal climate from turritellid gastropods. <i>6th International Sclerochronology Conference</i> , virtual, September 2022. (talk)
	Scholz, S.R. & S.V. Petersen. Reconstructing seasonal-scale precipitation patterns from high-resolution oxygen isotope sclerochronology of coastal turritellid gastropods. <i>GSA Connects</i> , virtual, October 2020. (talk)
	Scholz, S.R. , R. Seager, M. Ting & Y. Kushnir. Centennial timescale variability in Euro-Mediterranean hydroclimate since the Little Ice Age. <i>AGU Fall Meeting</i> , San Francisco, CA, December 2019. (poster)
	Scholz, S.R. , S.V. Petersen et al. Clumped isotope calibration for marine gastropods. <i>Michigan Geophysical Union</i> , Ann Arbor, MI, April 2019. (poster)
	Scholz, S.R. , S.V. Petersen et al. Isotope sclerochronology and tropical seasonality during the Mid-Miocene Climatic Optimum. <i>Goldschmidt</i> , Boston, MA, August 2018. (talk)

SOFTWARE	Arion — a climate-adaptive atmospheric river detection algorithm. • Performs consistently across a wide range of climates without adjustments or tuning. • Public catalogue to be released for MERRA-2 (1980–2025, hourly) and submitted to ARTMIP upon publication of the associated paper (currently in review at <i>Science Advances</i>). Island — improved custom land surfaces in Isca. • Allows creation of custom land surfaces with varying albedo, surface roughness, and bucket depths. • Linearized Penman-Monteith evaporation option for bucket model, with scalar or 2D fields of stomatal resistance and leaf area index to represent plants.																		
OTHER ACTIVITIES	<i>Rossbypalooza</i> summer school, University of Chicago, 2024 • Conducted research using convolutional neural nets to classify jet regimes relevant to heat extremes. • Gained experience with PyTorch, Keras, and unsupervised clustering techniques.																		
TEACHING	<table><tr><td>Teaching Fellow, EPS 1400 Atmosphere, Ocean and Climate Change</td><td>2025, 2026</td></tr><tr><td>Teaching Fellow, EPS 1010 Climate Change</td><td>2023</td></tr><tr><td>Teaching Fellow, EPS 1100 Dynamic Earth</td><td>2022</td></tr></table>	Teaching Fellow, EPS 1400 Atmosphere, Ocean and Climate Change	2025, 2026	Teaching Fellow, EPS 1010 Climate Change	2023	Teaching Fellow, EPS 1100 Dynamic Earth	2022												
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MENTORING	Jas Hollis (Yale undergraduate) • Undergraduate researcher examining the impact of different convective parameterizations on tropical circulation and energetics in Isca (primary advisor J.M. Lora). Yale EPS Graduate Peer Mentor (2025-present)																		
SERVICE	2026. Yale Pathways to Science summer enrichment program: “Plants: How Earth Became Green.” Three sessions for high school students. 2025. Session chair, “Atmospheric Rivers: Processes, Impacts, Observations, and Uncertainties,” <i>AGU Fall Meeting</i>. Reviewer: <i>Geophysical Research Letters</i>, <i>Journal of Geophysical Research: Atmospheres</i>, <i>Nature Communications</i>, <i>Communications Earth & Environment</i>, <i>Journal of Climate</i>, <i>Weather</i>.																		
AWARDS	<table><tr><td>Excellence in Teaching Prize, EPS 1400, Yale</td><td>2026</td></tr><tr><td>Hammer Prize, outstanding graduate student in Yale EPS</td><td>2025</td></tr><tr><td>Best Poster Presentation, Michigan Geophysical Union</td><td>2019</td></tr><tr><td>Turner Graduate Research Award, \$2000</td><td>2019</td></tr><tr><td>Undergraduate Academic Excellence Award, Earth and Environmental Science</td><td>2019</td></tr><tr><td>Eugene and Elizabeth Singer Award for Academic Excellence in Geology</td><td>2018</td></tr><tr><td>Matt Kelley/Granader Family Prize for Excellence in Writing</td><td>2017</td></tr><tr><td>Leonard and Eileen Newman Prize for Writing</td><td>2017</td></tr><tr><td>Roy W. Cowden Memorial Fellowship for Writing</td><td>2017</td></tr></table>	Excellence in Teaching Prize, EPS 1400, Yale	2026	Hammer Prize, outstanding graduate student in Yale EPS	2025	Best Poster Presentation, Michigan Geophysical Union	2019	Turner Graduate Research Award, \$2000	2019	Undergraduate Academic Excellence Award, Earth and Environmental Science	2019	Eugene and Elizabeth Singer Award for Academic Excellence in Geology	2018	Matt Kelley/Granader Family Prize for Excellence in Writing	2017	Leonard and Eileen Newman Prize for Writing	2017	Roy W. Cowden Memorial Fellowship for Writing	2017
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